On 'Ultrafilters with small generating sets'

Todor Dukov

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Introduction

The aim of this document is to give a better understanding of [1] by filling in details while also giving a high-level overview of the construction. We will talk about the ultrafilter number $\mathfrak{u}$ and the dominating number $\mathfrak{d}$. The main result of [1] is the consistency of

$$\mathfrak{u} = \kappa < \mathfrak{d} = \mathfrak{c} = \lambda$$

for any regular κ, λ, such that $\kappa < \lambda$.

The way we would do that is by first adding (with finite support iteration) λ-many Cohen reals and then very carefully adding κ-many Mathias reals. We shall treat the construction of the ultrafilter for Mathias with utmost care, so that

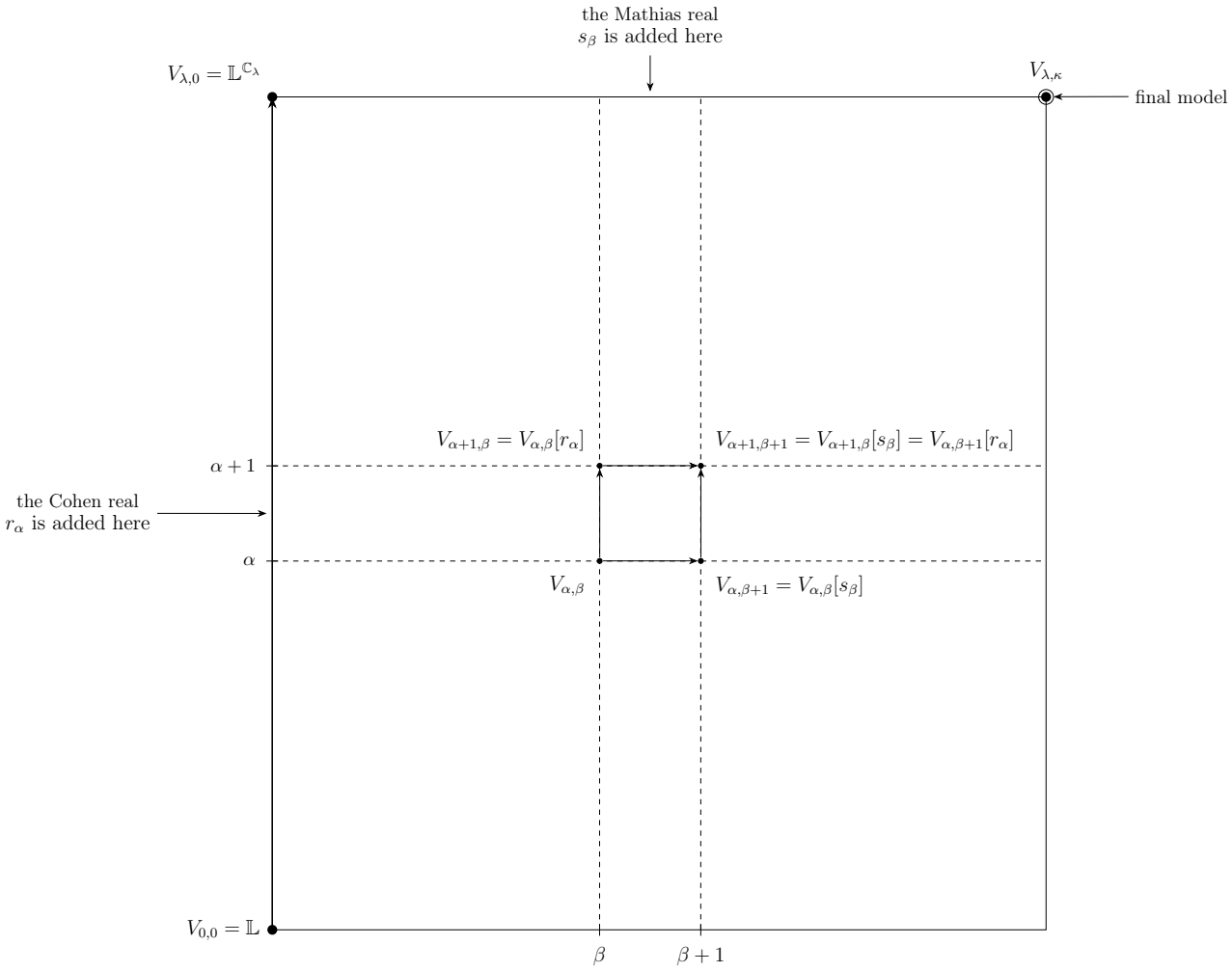
1. It does not ruin any work of the Cohen forcing for the dominating number.
2. The same Mathias real would be added no matter what level we are on the Cohen forcing iteration.

It is obvious why we have the first requirement. After Cohen forcing we would have $\mathfrak{d} = \mathfrak{c} = \lambda$. We would like to keep it that way during every step of the Mathias forcing. It is not so obvious why we need to satisfy the second requirement. If we satisfy the second requirement we would get what is called a matrix iteration. This would allow for more precise control of the model because we can attack the problem from two sides instead of one which would also help with the first requirement.

Let me try to explain it better. Suppose the added Cohen reals are $\{r_\alpha\}_{\alpha<\lambda}$ and the added Mathias reals are $\{s_\beta\}_{\beta<\kappa}$. For $\alpha < \lambda$ and $\beta < \kappa$ denote

$$V_{\alpha,\beta} = \mathbb{L}[\{r_\gamma\}_{\gamma<\alpha}][\{s_\delta\}_{\delta<\beta}].$$

What we would like is for $V_{\alpha+1,\beta+1}$ to be related to both $V_{\alpha,\beta+1}$ and $V_{\alpha+1,\beta}$. Being related to both would guarantee the niceties of both our Mathias forcing and Cohen forcing. This is how it looks visually:



The method was introduced in [1]. It was further developed in [2], where the terminology *matrix iteration* appeared for the first time. Since first introduced, it has proven very useful in tackling combinatorial problems concerning cardinal characteristics of the reals.

So we talked about what we want for $\mathfrak{d}$, but what about $\mathfrak{u}$? Why would we have $\mathfrak{u} = \kappa$ in the end? Turns just having each ultrafilter for stage contain the currently added Mathias reals is enough, as we shall see next.

Why $\mathfrak{u} = \kappa$?

We will show this in two directions. First we will show an ultrafilter base of size κ. Then we will prove that no family of size smaller than κ forms an ultrafilter base.

Let $\mathcal{U}_\beta$ be the ultrafilter in step $\beta < \kappa$ of Mathias forcing. If for each $\beta < \kappa$, the ultrafilter $\mathcal{U}_\beta$ contains $\{s_\delta\}_{\delta<\beta}$, the family $\{s_\beta \setminus x \mid \beta < \kappa \ \& \ x \text{ is finite}\}$ will form a base for an ultrafilter. Let $A \in [\omega]^\omega \cap V_{\lambda,\kappa}$. Then since κ is regular, $A \in [\omega]^\omega \cap V_{\lambda,\beta}$ for some $\beta < \kappa$. This is true because in finite support iterations, limit stages of regular cofinality add no new reals. By properties of Mathias forcing, either A or $\omega \setminus A$ almost includes s_β. Therefore, $\mathfrak{u} \leq \kappa$.

We will show that $\kappa \leq \mathfrak{u}$. Let $\mathcal{A}$ be a family of subsets of ω in $V_{\lambda,\kappa}$ of size less than κ. By similar arguments, $\mathcal{A} \in V_{\lambda,\beta}$ for some $\beta < \kappa$. We will show a set $X \in V_{\lambda,\kappa}$ such that neither X nor $\omega \setminus X$ includes a member of $\mathcal{A}$. Define

$$X = \{n \in \omega \mid |n \cap s_\beta| \text{ is even}\}.$$

Assume that either X or $\omega \setminus X$ includes some $A \in \mathcal{A}$. Let (z, Z) be a condition forcing that. Let $m = \min Z$ and let a be an element of A that is larger than m. Then

- $(z, Z \setminus a)$ is an extension of (z, Z) forcing $a \cap s_\beta = z$.
- $(z \cup \{m\}, Z \setminus a)$ is an extension of (z, Z) forcing $a \cap s_\beta = z \cup \{m\}$.

One of the conditions force $a \in X$ and the other $a \notin X$, which is impossible, since

- If (z, Z) forces $A \subseteq X$, then $a \notin X$ can't be forced by any extension of (z, Z).
- If (z, Z) forces $A \subseteq \omega \setminus X$, then $a \in X$ can't be forced by any extension of (z, Z).

We have proved that neither X nor $\omega \setminus X$ includes a member of $\mathcal{A}$, so $\mathcal{A}$ can't form an ultrafilter base.

Construction of the ultrafilters

T.B.D.

References

[1] Blass, A., and Shelah, S. Ultrafilters with small generating sets. *Israel Journal of Mathematics 65*, 3 (1989), 259–271.

[2] Brendle, J., and Fischer, V. Mad families, splitting families and large continuum. *The Journal of Symbolic Logic 76*, 1 (2011), 198–208.